

FARAAZ MOHAMMED

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Temporary Graduate visa (subclass 485) — full Australian work rights, no sponsorship required • Available immediately

SUMMARY

Master of Artificial Intelligence graduate (RMIT, Jul 2026) with a B.Tech in Information Technology, working across applied ML and software engineering. At Prodegee, designed the school-geolocation algorithm at the core of a country-scale electrification pipeline (7.2M building polygons, 18,121 schools) and the sensitivity analysis that isolated the parameter behind a 90% household under-count. Co-author of a peer-reviewed computer-vision paper (94.75% accuracy). In 2026 independently shipped a Kotlin Android app, an offline-first PWA and an MCP-based local-LLM delegation system — requirements first, reviewed, unit-tested. Open to roles anywhere in Australia.

EXPERIENCE

AI/ML Engineering Intern — Prodegee

Mar 2026 – Jun 2026

Remote, Melbourne • RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Designed and built the three-stage school-geolocation refinement algorithm that became the team's core pipeline: a national layer of 7.2M Overture Maps building polygons (GeoPandas/Shapely → GeoPackage, wrapped in a Kestra flow), BFS spatial clustering with composite scoring, and a nearest-school proximity check; ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Made scoring measurable and portable: replaced a binary distance cutoff with exponential decay, designed a weighted per-correction confidence score with weights derived from measured fix attribution, and added sigmoid normalisation with per-country inflection points so one threshold works across all three countries; authored the cross-team handoff contracts.
- Built a $\pm 20\%$ sensitivity harness that ranked cluster radius as the dominant parameter (± 36 – 44% swing in all 191 villages), driving the 75 → 200 m recalibration that lifted buildings-per-household from 0.069 (a 90%+ under-count) to 0.415, into the target band.
- Built an IoU harness for compound polygons and showed that all three generators failed the ≥ 0.5 gate because the upstream cluster was $\sim 35\times$ over-scoped — a scoping fault, not a fitting one — which redirected the next assignment.
- Ran footprint inference across 200 villages in 7 Mauritanian Wilayas, implemented the Multi-Tier Framework tier heuristic and an XGBoost multi-output tier regressor, and documented that the regressor's 0.816 "confidence" measured agreement with the heuristic it was trained on, not accuracy.
- Corrected a published baseline (6–13% → 37%) against a co-built 40-school satellite-verified ground-truth set, and published a negative result — a proximity penalty hurt accuracy at all three thresholds — closing off a dead parameter.
- Led 3 of 10 assignments; primary author of 3 of 4 formal deliverables; 68 commits, 7 peer reviews; pinned upstream inputs by commit SHA for reproducible re-runs; wrote the semester-2 handoff memo. Built `model_router.py`, a cost-proportional Claude API router (cached Haiku classifier → Haiku / Sonnet / Opus).

Tools: Python, GeoPandas, Shapely, scikit-learn, XGBoost, pandas, Kestra, QGIS, GeoPackage, Git, Claude Code (MCP, hooks)

Dispatch Team Member — Amart Furniture

Feb 2025 – Mar 2026

Melbourne, VIC • part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory-management system, processed customer order workflows, coordinated real-time fulfilment and resolved order discrepancies across 5 Melbourne stores; trained a new team member.

EDUCATION

Master of Artificial Intelligence — RMIT University

Jul 2024 – Jul 2026

Melbourne • Coursework: Intelligent Decision Making (ASP/Clingo), Computer Vision, Machine Learning, AI Systems Design

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology

Dec 2020 – May 2024

KMIT, Hyderabad, India (affiliated to JNTUH) • Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

KEY PROJECTS

Time Flow — ADHD-focused planner (PWA)

JavaScript, Service Worker, Web Notifications, Kotlin bridge

Jul – Sep 2026

- Built and deployed an installable, offline-first PWA (4,600 lines of vanilla JS, no framework or build step) that packs goals, recurring targets and reminders into a user's free minutes using an urgency $\times$ importance scoring model, with period-aware urgency for recurring quotas.
- Implemented dual-timer focus sessions with auto-advance, weekly/monthly reminder scheduling and a cache-first service worker delivering system notifications; designed a JS ↔ Kotlin @JavascriptInterface bridge so the same code runs in a native Android wrapper with exact alarms; fixed input-focus loss under 1 s live re-renders with a targeted DOM-patch path.

↳ Live: faraaz817.github.io/time-flow-web • GitHub: github.com/faraaz817/time-flow-web

MGM Prayer Alerts — Android app for Melbourne Grand Mosque · Kotlin, Android 8–15, AlarmManager, JUnit Aug 2026

- Shipped a community app that fetches the mosque's daily timetable, schedules exact Azaan alarms with per-prayer toggles and custom notification-channel sounds, refreshes nightly and survives reboot; sideloaded to community members.
- Wrote the functional requirements (scope, decision log, acceptance criteria) before coding, then ran a severity-ranked code review against them: found and fixed a critical 12-hour AM/PM storage bug and a one-shot refresh alarm that stopped re-arming after a failed fetch; added round-trip unit tests for time parsing.

↳ GitHub: github.com/faraaz817/MgmPrayerAlerts

Gemma Delegation System — local/cloud LLM hybrid · Node.js, MCP SDK, Ollama, Unsloth/QLoRA, Colab May 2026

- Built an MCP server through which Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma 4 E2B, failing open to the frontier model whenever Ollama is unavailable so delegation never blocks the user.
- Closed the loop: accept/reject feedback updates a capability profile that governs future delegation and curates a JSONL dataset; a Colab notebook fine-tunes on confirmed examples (4-bit QLoRA, LoRA r=16) and exports GGUF back into Ollama as a drop-in replacement.

↳ GitHub: github.com/faraaz817/gemma-delegation-system

- **Bro — interactive AI portfolio chatbot:** recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; Anthropic API, deployed on Vercel. (Next.js, TypeScript, Prisma/Postgres)
↳ Live: job-go-mauve.vercel.app/portfolio
- **Fire Detection & Alerting System — B.Tech capstone (team of 4):** trained a custom Haar cascade in OpenCV for real-time webcam fire detection, chosen over heavier detectors for latency; on detection a threaded pipeline plays an audible alarm and dispatches WhatsApp alerts (RAPIWHA API) to the owner and nearest fire station without blocking the capture loop. (Python, OpenCV, threading, requests)
- **NeuroVisualZ — exam-timetabling AI, RMIT (team of 3):** built the YAML-to-ASP parser/encoder translating timetabling instances into Clingo logic programs; scoped Levels 1–4 (room allocation, ITC'07 soft constraints, SWI-Prolog validator). (Python, Clingo/ASP, SWI-Prolog)
- **Agent skills for Claude Code / Cursor:** authored five reusable skills — CO-STAR requirements builder, Socratic tutor, better-way reviewer, skill sync, and a pikepdf CLI that unlocks signature-locked PDFs — each a documented, trigger-scoped procedure. github.com/faraaz817/My_Skills
- **IBM Z development toolchain:** connected VS Code to a live z/OS environment via Zowe CLI/Explorer and z/OSMF REST APIs; submitted and verified a JCL job with condition code 0000. (IBM z/OS, Zowe, JCL)

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. “A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning.” *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Benchmarked 5 edge-detection operators × 8 classifiers (40 combinations) as a lesion feature-extraction step on a 15,000-image, four-class maize dataset; best result Sobel + MLP at 94.75%. Implemented the Canny arm (OpenCV + scikit-learn, ten classifiers). ↳ questjournals.org/jses · Code: github.com/faraaz817/Machine_Learning

TECHNICAL SKILLS

Languages: Python (primary), Kotlin, JavaScript, TypeScript, SQL, Prolog / ASP (Clingo), JCL, Bash

AI / ML: scikit-learn, XGBoost, OpenCV (image classification, edge detection, Haar cascades), evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction), LLM fine-tuning (Unsloth, QLoRA, GGUF), Ollama, Answer Set Programming, prompt engineering, retrieval-grounded generation

Geospatial: GeoPandas, Shapely, QGIS, PostGIS, Overture Maps, OpenStreetMap, GeoPackage / GeoJSON / Parquet

Software: Android (Kotlin, AlarmManager, BroadcastReceiver, notification channels, Gradle KTS, JUnit), PWA (service workers, Web Notifications), Node.js, Next.js, Prisma, REST/JSON integration

AI-assisted engineering: MCP servers, Claude Code (skills, hooks, subagents), Cursor, Anthropic API (prompt caching, routing)

Tooling & platforms: Git / GitHub, Kestra, Docker (basic), Colab / Jupyter, Vercel, GitHub Pages, IBM z/OS + Zowe, Linux

CERTIFICATIONS, AWARDS & LEADERSHIP

- IBM Z Xplore – Concepts and IBM Z & LinuxONE Community Contributor – Level 1 (IBM digital badges via Credly, 2026).
- Smart India Hackathon delegate, Government of India (2022) · Certificate of Publication, Quest Journals (2023).
- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.
- Volunteer fundraiser, Taher Foundation (2023).

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)